Ficking Alindra Wijayadarma

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Portfolio: https://canva.link/quvcclw7e7auh99

A Cartography and Remote Sensing graduate from Universitas Gadjah Mada with a GPA of 3.81/4.00. Possesses competencies and a strong interest in Geographic Information Systems (GIS), remote sensing, and geospatial analysis to support planning and decision-making in urban development, plantation management, environmental studies, and mining. Skilled in spatial data processing, analysis, and visualization. Demonstrates strong coordination, communication, and teamwork skills, with a commitment to continuous professional development and contributing to collaborative projects through the application of geospatial technologies.

EDUCATION

Universitas Gadjah Mada (UGM), Faculty of Geography Aug 2022 - Sep 2026
Cartography and Remote Sensing | GPA: 3.81/4.00

- Thesis: Spatial Analysis of Traffic Accident Vulnerability Using Aerial Photo and a Weighted Overlay Quantitative Approach in Magelang City.

PROFESSIONAL EXPERIENCE

Faculty of Geography, UGM Oct 2025 - Oct 2026
Research Assistant, Parking Capacity Estimation in UGM Campus Parking Pockets

- Supported an independent research project on estimating parking capacity in the campus parking pockets of Universitas Gadjah Mada using high-resolution UAV aerial photographs.
- Digitized parking areas visually from aerial photographs to measure total parking area (m^2) as the basis for capacity estimation.
- Applied spatial calculation of capacity from parking area, space required per vehicle, and circulation space, and estimated parking availability to support data-based campus parking management.

DPUP-ESDM Yogyakarta Aug 2025 - Oct 2025
Intern

- Consolidated 500+ spatial records on mining business permits (IUP), people's mining permits (IPR), mineral potential, and quarry sites, improving data reliability and speeding up permit verification by approximately 30%.
- Digitized and georeferenced 150+ mining maps, improving boundary accuracy (±5 m) to support compliance monitoring and reduce spatial discrepancies in regulatory oversight.
- Supported development of the 2025 Regional Mining Area Map by integrating updated permit and resource data, achieving 100% alignment of active mining permits with provincial spatial planning priorities.

Department of Geographic Information Science, UGM Feb 2025 - May 2025
Laboratory Teaching Assistant, Geographic Information Systems

- Guided 13 undergraduate students through GIS lab practicals (georeferencing, digitization, geoprocessing, symbology, and cartographic layout), with mentored students' average score above the practicum average.
- Co-developed lab modules on spatial analysis and geospatial visualization to minimize common technical errors.
- Supported student spatial projects from data processing to final map production, with 100% of assessed outputs completed.

Department of Geographic Information Science, UGM Aug 2024 - Dec 2024
Laboratory Teaching Assistant & Practicum Coordinator, Algorithms and Programming

- Taught algorithm design and computational problem-solving to 11 students, with trained students' average score above the practicum average.
- Provided hands-on guidance in debugging and code optimization for geospatial applications.
- Designed programming modules for geospatial problem-solving and completed 100% of practicum milestones within the semester.

Transport Studies Community (TRANSCO) Apr 2022 - May 2022
Intern

- Conducted peak-hour vehicle counts across road segments in the Bogor metropolitan area, providing input for traffic flow and road capacity evaluation.
- Processed and validated 1,000+ traffic observation records, reducing data input errors and inconsistencies by 25%.
- Assisted with primary data collection and preliminary analysis, contributing to baseline datasets covering 100% of targeted study corridors.

ACADEMIC FIELD PROJECTS

Field Course III - West Bali, UGM Dec 2024 - Jul 2025
Chairperson

- Led and coordinated 98 students across 8 multidisciplinary research themes, ensuring all outputs were completed on time.
- Oversaw the project from pre-field planning through fieldwork and data synthesis, delivering 100% of milestones on schedule and aligning research methods across thematic clusters.

Research Theme: Suitability Analysis of Regional Spatial Plan (RTRW) Against Settlement Land Use Using Remote Sensing - Negara and Jembrana Districts, Jembrana Regency

- Acquired and processed DEM, remote sensing imagery, and existing maps as base data for spatial suitability analysis.
- Conducted field validation and interviews at 40+ sample points to verify land use and physical conditions.
- Developed thematic maps (soil texture, slope, drainage, groundwater basin, flood hazard, land use) and performed overlay analysis to produce land suitability maps against the RTRW.

Field Course II - Kebumen, UGM Oct 2023 - Jul 2024

Team Leader, Large-Scale Mapping Research Theme

- Led 18 students in large-scale spatial research on land use, socio-economic dynamics, and solid waste management, completing all field and analytical outputs within the academic timeline.
- Directed the research from field survey design and primary data collection to GIS-based spatial analysis, and oversaw the final research report.

Sub-theme: Land Use Mapping

- Digitized land use on screen from high-resolution remote sensing imagery and produced pre-field, reinterpretation, and final maps.
- Conducted ground-truthing at 50 sample points, including density measurements for 15 settlement blocks.

ORGANIZATIONAL EXPERIENCE

Geographic Information Science Student Association (HMSaIG), UGM Mar 2023 - May 2025

Staff, Talent and Interest Division

- Designed and ran sports and arts programs for 300+ students across three cohorts, increasing active participation by approximately 35% year over year and sustaining 90% attendance at major events.
- Built and maintained a member database of 300+ profiles to support coordination and talent development.

Pionir Geospace, UGM May 2024 - Feb 2025

Public Relations Division Staff

- Handled external communications and multi-platform campaigns, producing 30+ promotional contents (posters, videos, copywriting) that increased event engagement by approximately 40%.
- Managed institutional social media channels, improving reach and interaction by approximately 35%.
- Served as liaison officer for 50+ invited speakers, faculty members, and keynote guests, coordinating their attendance.

GIS Competition (GISCOM & MAPCOM), UGM Apr 2023 - Sep 2023

Secretariat Division Staff

- Managed documentation for 39 competition teams, keeping participant records 100% complete.
- Validated and cross-checked administrative data, reducing documentation errors by approximately 30%.
- Prepared administrative reports, finalist certificates, and official archives before the competition deadline.

CERTIFICATIONS AND TRAINING

Spatial Data Science and Applications | *Yonsei University* Oct 2026

GIS: Geographic Information Systems for Sustainability | *University of Michigan* Oct 2026

Occupational Health and Safety Management System (OHSMS) Based on SNI ISO 45001:2018 Oct 2026
National Standardization Agency of Indonesia (BSN)

Risk Management Based on SNI ISO 31000 | *National Standardization Agency of Indonesia (BSN)* Oct 2026

Python Essentials 1 Oct 2026
Cisco Networking Academy, in collaboration with OpenEDG Python Institute

Data Science Essentials with Python | *Cisco Networking Academy* Oct 2026

Masterclass GIS for Mining: ArcGIS Pro | *Takeclass.id* Apr 2026

CORE COMPETENCIES

GIS & Remote Sensing Software: ArcGIS Pro, ArcGIS Desktop/ArcMap, ArcScene, QGIS, ENVI, SNAP, Google Earth Engine, ILWIS.
Technical Skills: Geographic Information Systems (GIS), Remote Sensing, Urban Remote Sensing, Spatial Data Analysis & Visualization, Georeferencing & Digitization, Geoprocessing, Overlay & Land Suitability Analysis, Thematic & Cartographic Mapping, Field Survey & Ground-Truthing, Traffic Data Collection & Enumeration.
Other Tools: Microsoft Office (Word, Excel, PowerPoint), Adobe Premiere, Visual Studio Code, Blender, R Studio.
Leadership & Professional Skills: Team Leadership & Coordination, Project Management, Mentoring & Training, Analytical Thinking, Stakeholder Liaison, Adaptability.

LANGUAGES

Indonesian (Native) | English (Intermediate)